

# Robotic System for Monitoring Industrial Installations

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## 1. Introduction

The inspection of industrial electrical infrastructure exposes technicians to significant risks, including electrocution and arc flash. Safety constraints limit how close workers can get to live equipment, while logistical costs restrict how frequently inspections can be carried out. Thermal imaging is a well-established method for detecting early signs of equipment failure, but its deployment at scale remains constrained by the same human factors.

## 2. Motivation

By integrating a thermal camera into a quadruped robot, inspections can become remote, repeatable and safer, while enabling more frequent monitoring of critical infrastructure. The *Unitree Go2*, costing around €1,300–€10,000, offers a much lower-cost alternative to platforms such as *Boston Dynamics' Spot* (60.000€) or *Anybotics' Anymal* (130.000€), making robotic thermal inspection more accessible.

## 3. System Architecture

The system (**Figure 2**) is built around a web-based operator interface (**Figure 3**) through which the user can control the *Go2*, visualize its *RGB* camera and *LiDAR* point cloud with the addition of the thermal feed and control of the gimbal. The robot is controlled over *WebRTC*, while a *Raspberry Pi 4* mounted on the robot (**Figure 1**) handles the thermal camera and gimbal control through *WebRTC* and *Websockets* respectively.

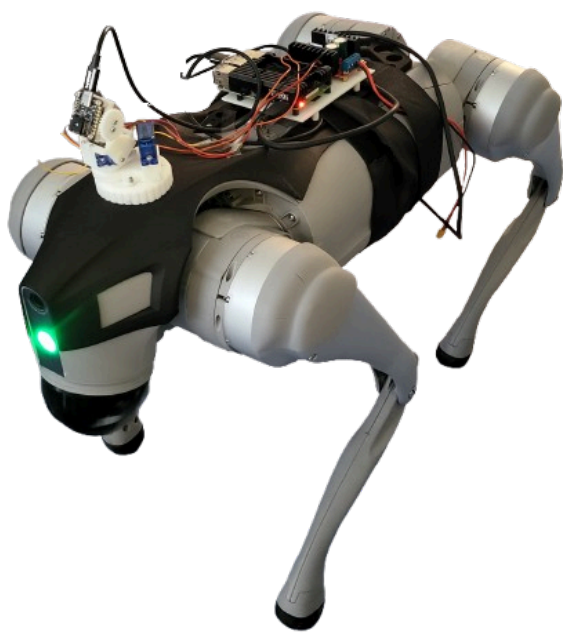


Figure 1: Finished project

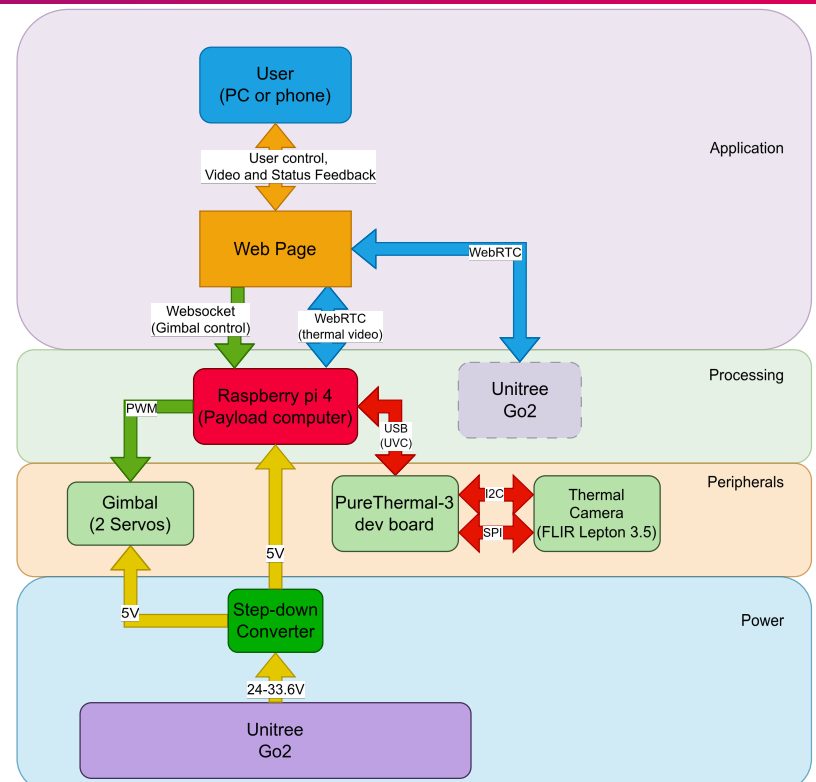


Figure 2: System Logic Block Diagram

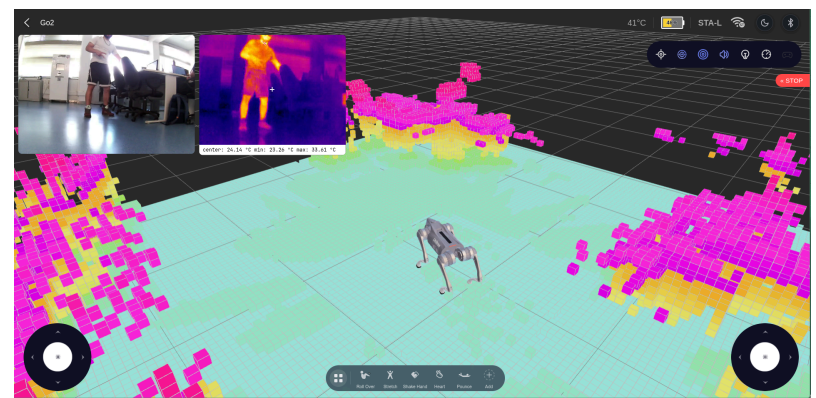


Figure 3: Web control interface

## 4. Results & Conclusion

The system was validated at a transformer substation at ISEL, where two active electrical faults were successfully detected. The first, an oxidized cable (**Figure 4a**), reached a peak temperature of 184°C, exceeding the 153°C measurement ceiling of the professional handheld camera used as reference.

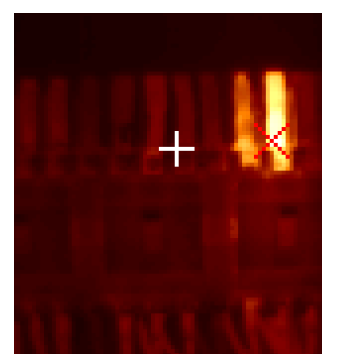
Both faults were correctly identified and localized despite the sensor's limited 160×120 pixel resolution, demonstrating that the system is capable of detecting thermally significant faults in real industrial environments at a fraction of the cost of existing commercial solutions.



(a)



(b)



(c)

Figure 4: Thermal inspection of an oxidized cable